Kernix 0.9.0 / Performance brief

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

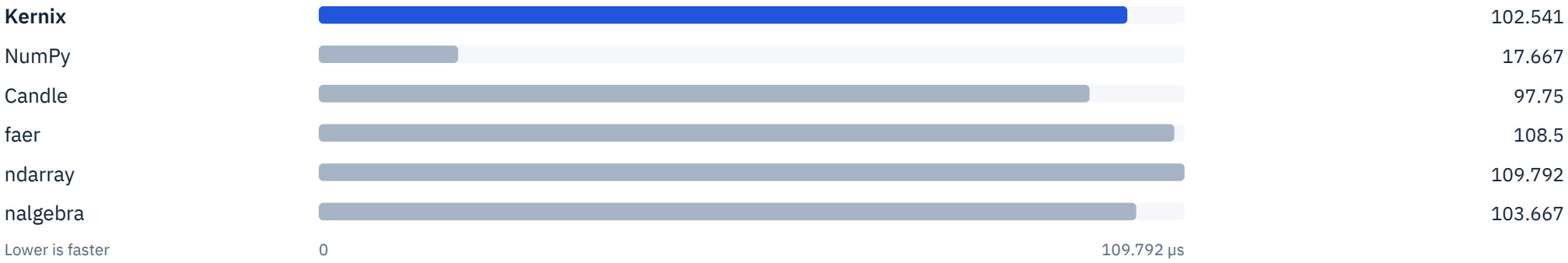
25 workloads

100 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

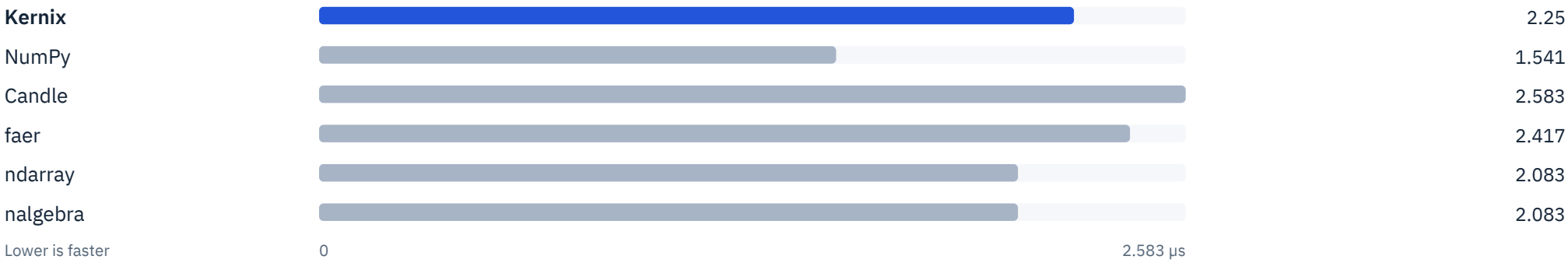
Kernix 0.9.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

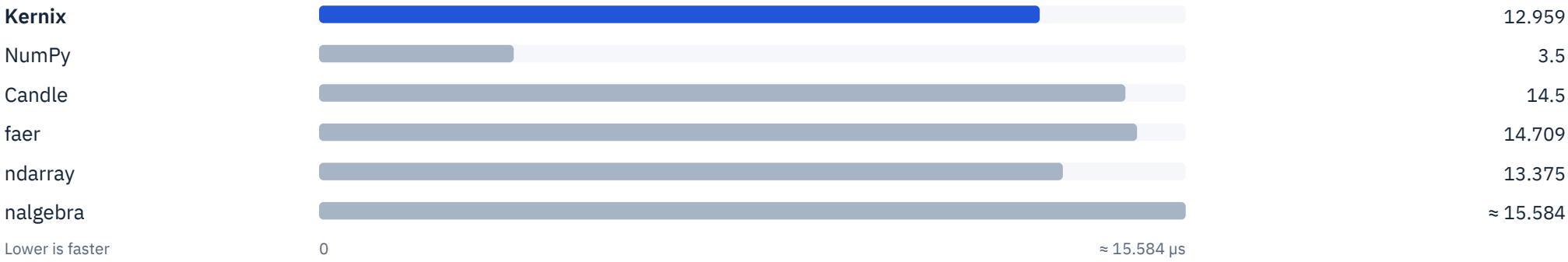
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



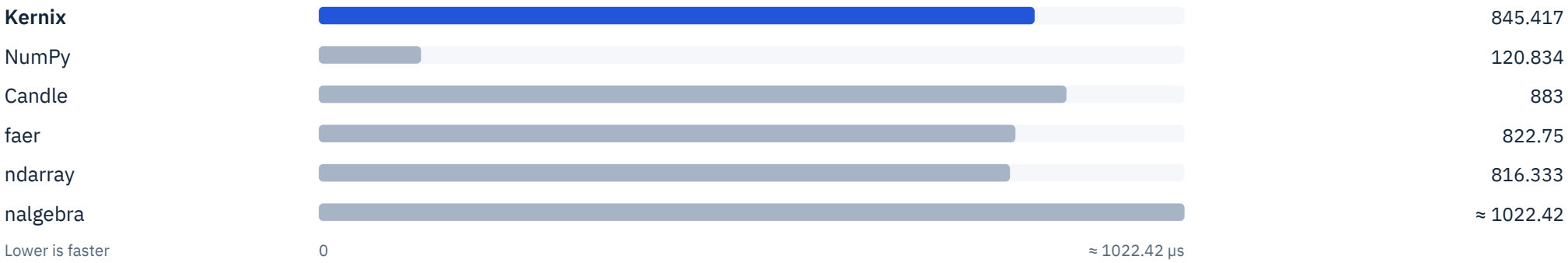
Kernix 0.9.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

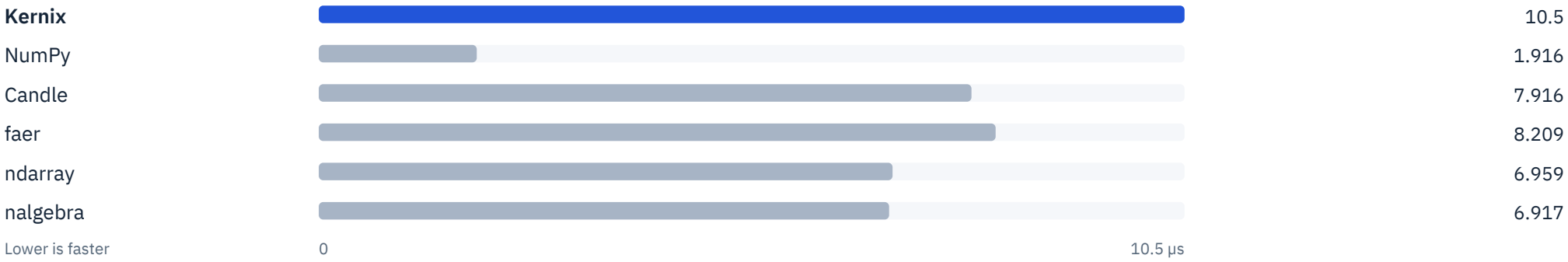
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



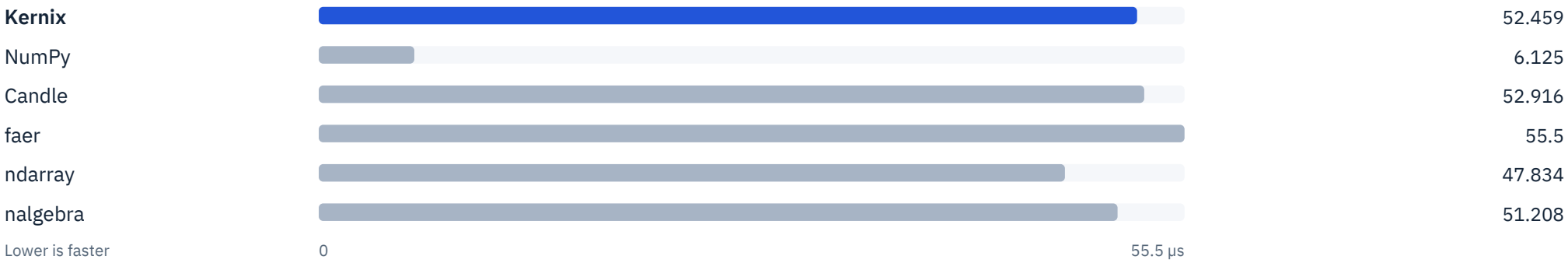
Kernix 0.9.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

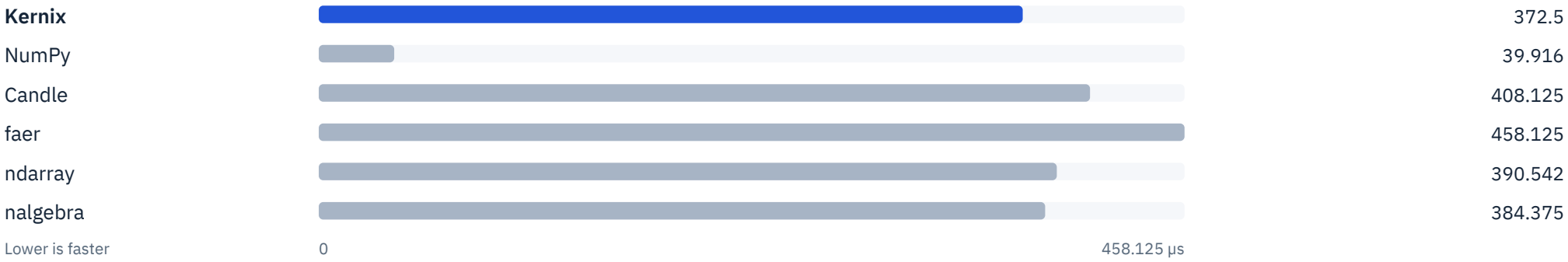
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product

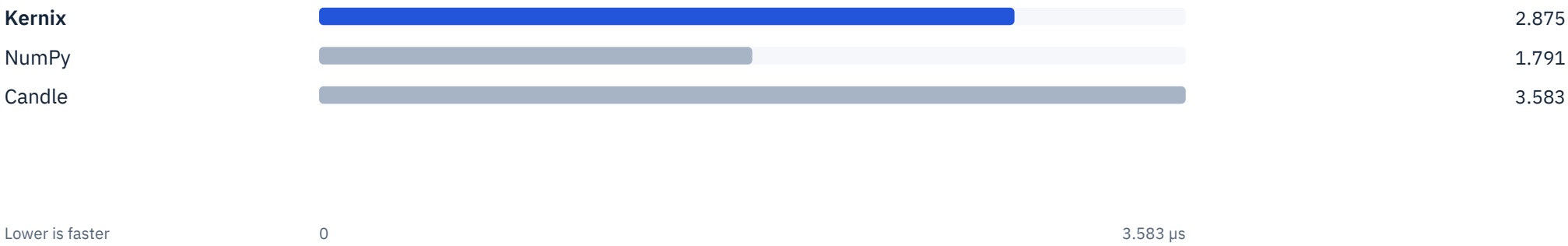


Kernix 0.9.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



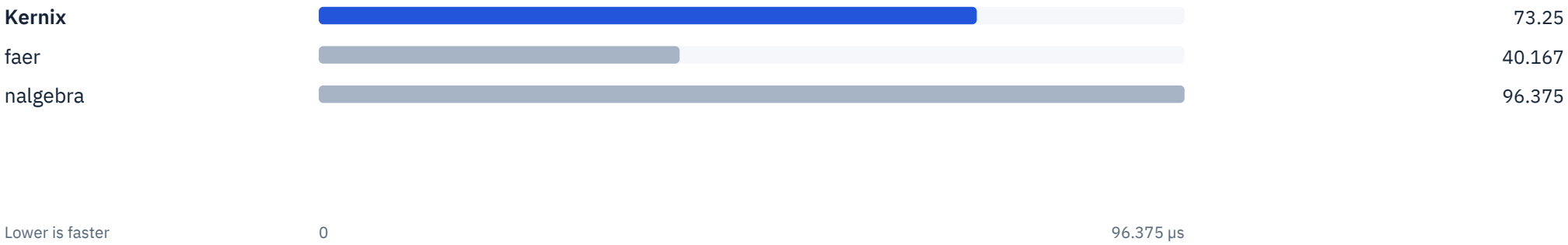
Kernix 0.9.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Column-pivoted QR

Per-call median

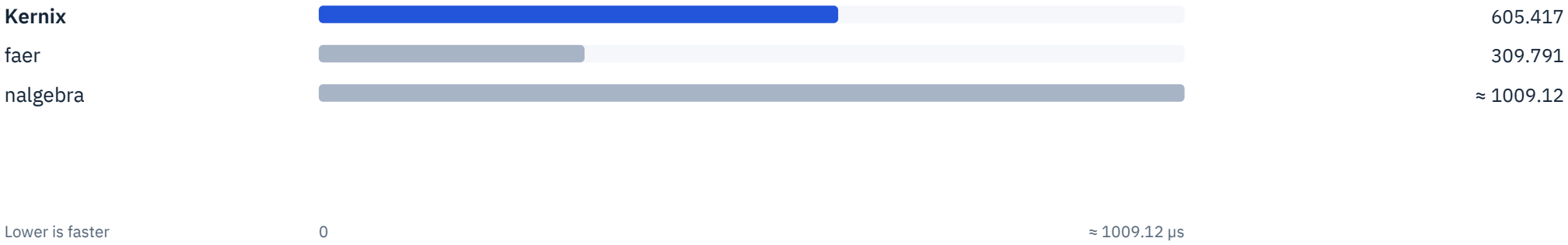
96 × 32 · f64 · 1 worker · allocated output



Column-pivoted QR

Per-call median

256 × 64 · f64 · 1 worker · allocated output



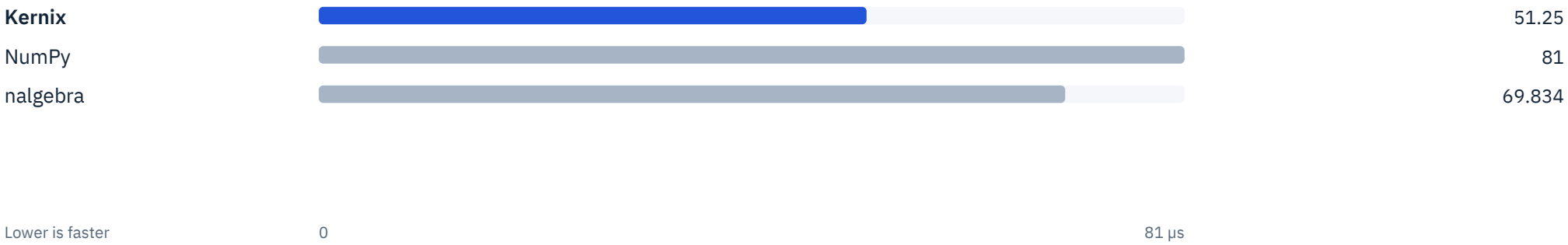
Kernix 0.9.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Minimum-norm least squares

Per-call median

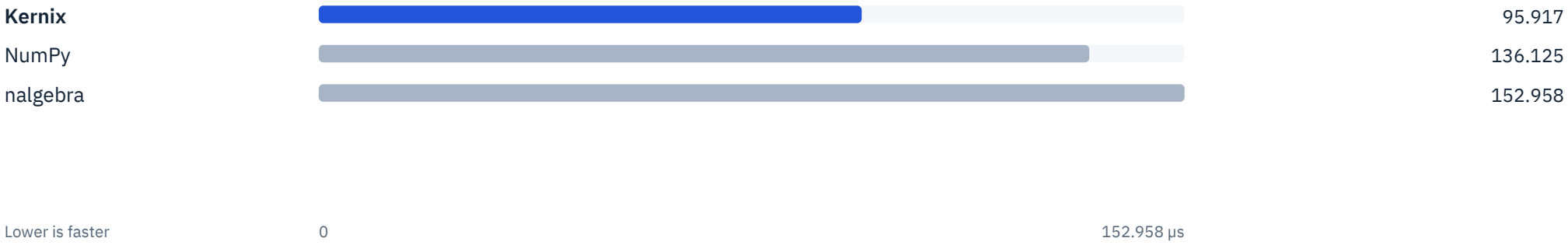
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

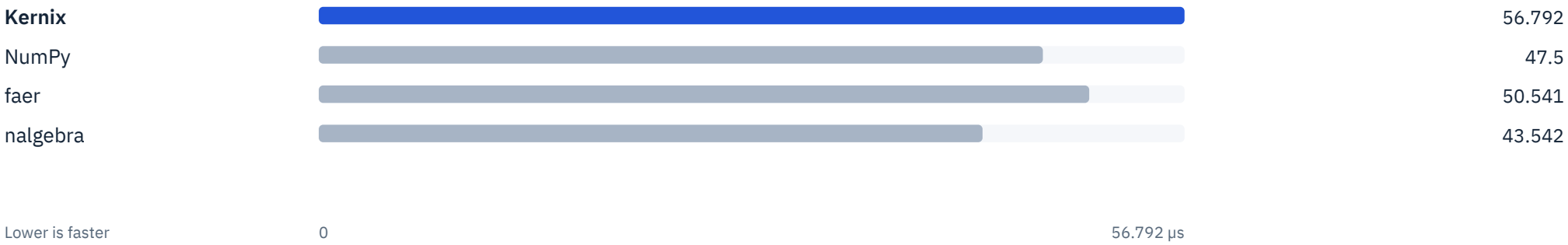


Kernix 0.9.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

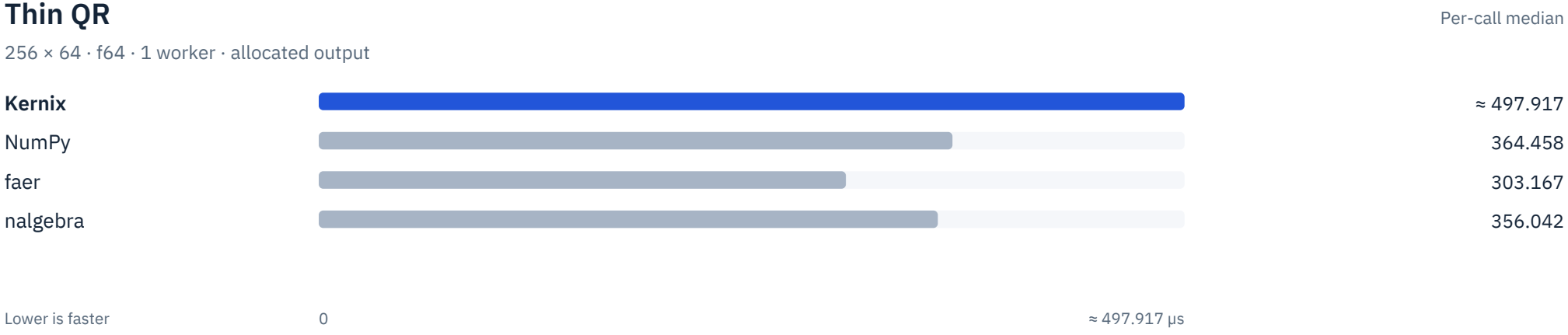
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output

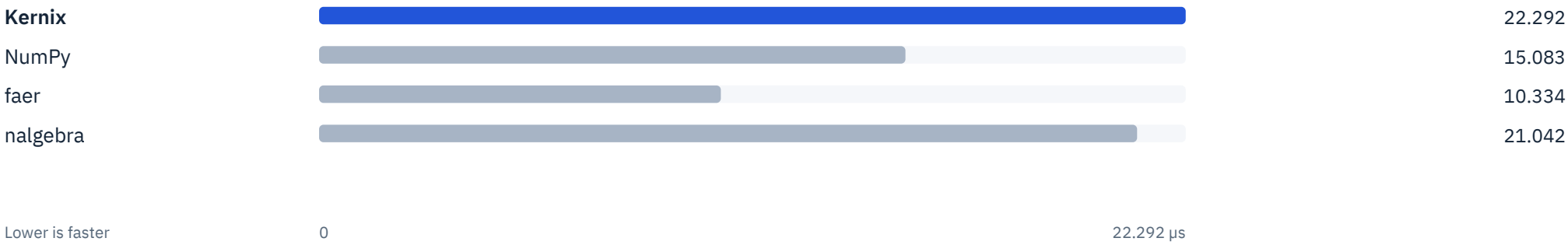


Kernix 0.9.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

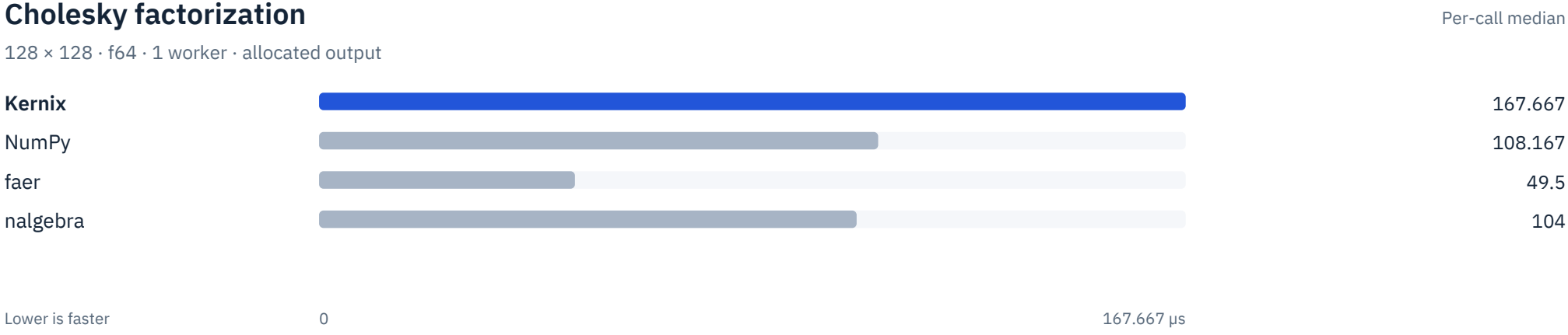
Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output



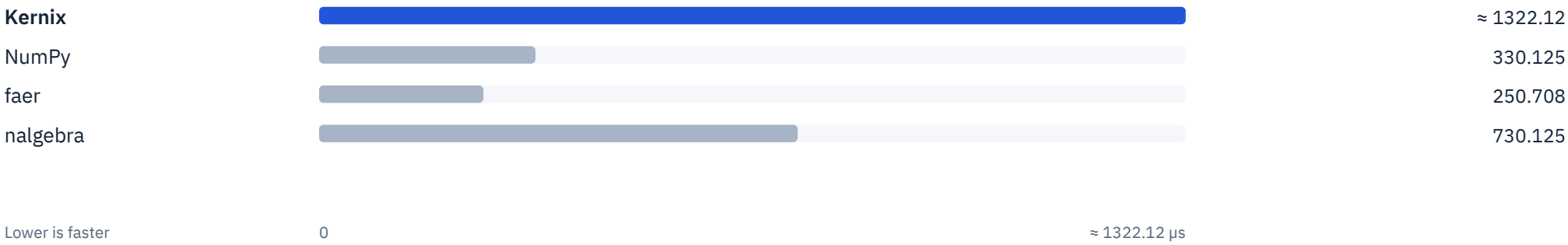
Kernix 0.9.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.9.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Record + backward

Per-call median

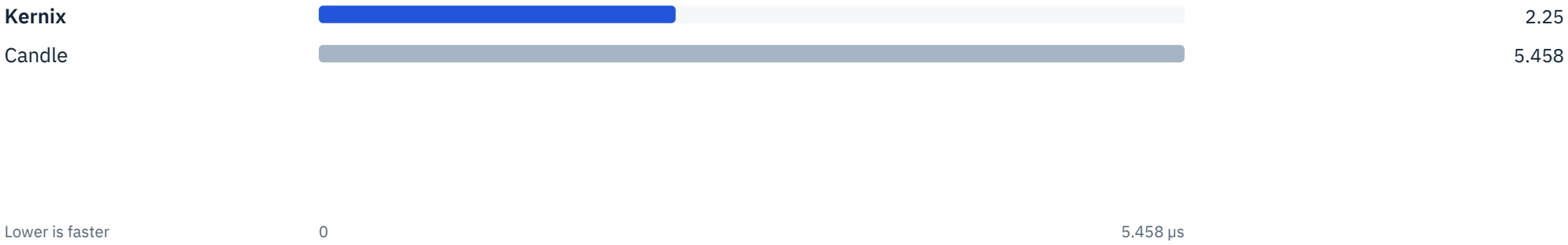
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.9.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model recording + forward

Per-call median

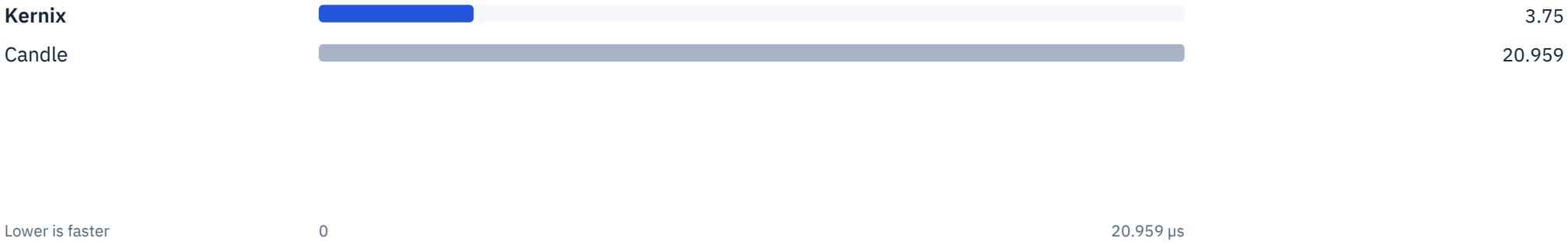
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.9.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Parameter update

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

Per-call median

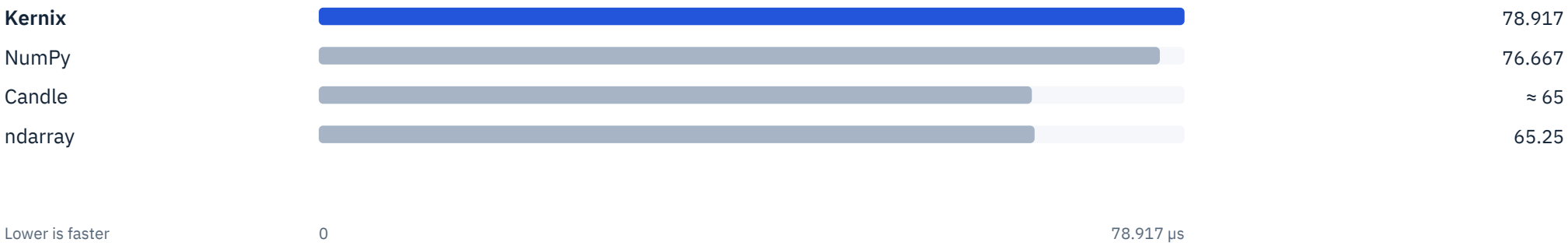


Kernix 0.9.0 / Statistics & activations

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Row softmax

256 × 64 · f64 · 1 worker · allocated output



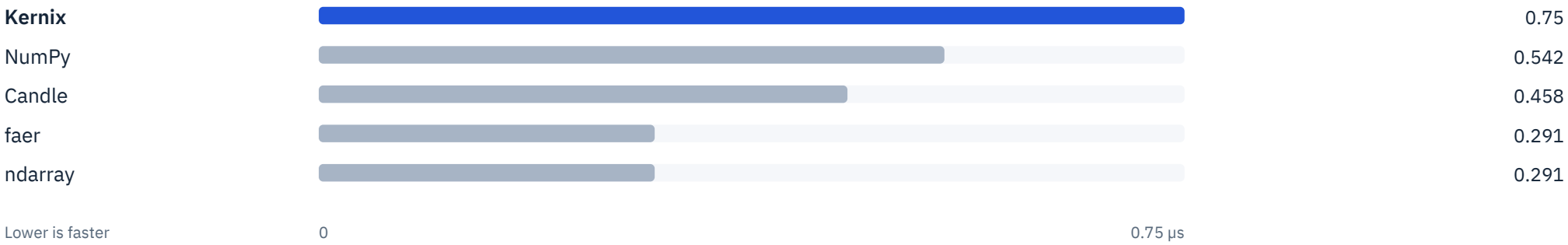
Kernix 0.9.0 / Tensor construction

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2250	2.25
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2417	2.417
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2083	2.083
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2583	2.583
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	1541	1.541
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2083	2.083
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	3500	3.5
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	15583.999999999998	15.583999999999998
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	12959	12.959
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	13375	13.375

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14709	14.709
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	14500	14.5
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	108500	108.5
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	17667	17.667
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	103667	103.667
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	109792	109.792
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	102541	102.541
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	97750	97.75
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	1022417.0000000001	1022.4170000000001
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	120834	120.834

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	845417	845.417
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	883000	883
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	822750	822.75
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	816333	816.333
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6917	6.917
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	1916	1.916
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7916	7.916
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	10500	10.5
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	8209	8.209
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6959	6.959

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	52459	52.459
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	6125	6.125
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	51208	51.208
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	52916	52.916
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	47834	47.834
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	55500	55.5
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	39916	39.916
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	458125	458.125
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	372500	372.5
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	384375	384.375

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	408125	408.125
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	390542	390.542
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3583	3.583
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	1791	1.791
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2875	2.875
Column-pivoted QR / faer 96 × 32 · f64	40167	40.167
Column-pivoted QR / Kernix 96 × 32 · f64	73250	73.25
Column-pivoted QR / nalgebra 96 × 32 · f64	96375	96.375
Column-pivoted QR / nalgebra 256 × 64 · f64	1009125	1009.125
Column-pivoted QR / faer 256 × 64 · f64	309791	309.791

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / Kernix 256 × 64 · f64	605417	605.417
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	51250	51.25
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	81000	81
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	69834	69.834
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	95917	95.917
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	136125	136.125
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	152958	152.958
Thin QR / faer 96 × 32 · f64	50541	50.541
Thin QR / NumPy 96 × 32 · f64	47500	47.5
Thin QR / nalgebra 96 × 32 · f64	43542	43.542

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / Kernix 96 × 32 · f64	56792	56.792
Thin QR / faer 256 × 64 · f64	303167	303.167
Thin QR / Kernix 256 × 64 · f64	497916.99999999994	497.91699999999994
Thin QR / nalgebra 256 × 64 · f64	356042	356.042
Thin QR / NumPy 256 × 64 · f64	364458	364.458
Cholesky factorization / Kernix 64 × 64 · f64	22292	22.292
Cholesky factorization / nalgebra 64 × 64 · f64	21042	21.042
Cholesky factorization / faer 64 × 64 · f64	10334	10.334
Cholesky factorization / NumPy 64 × 64 · f64	15083	15.083
Cholesky factorization / faer 128 × 128 · f64	49500	49.5

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	167667	167.667
Cholesky factorization / NumPy 128 × 128 · f64	108167	108.167
Cholesky factorization / nalgebra 128 × 128 · f64	104000	104
Cholesky factorization / nalgebra 256 × 256 · f64	730125	730.125
Cholesky factorization / faer 256 × 256 · f64	250708	250.708
Cholesky factorization / Kernix 256 × 256 · f64	1322125	1322.125
Cholesky factorization / NumPy 256 × 256 · f64	330125	330.125
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	8042	8.042
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	19416	19.416
Model inference / Kernix batched normalized/sliced MLP · f64	2250	2.25

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Candle batched normalized/sliced MLP · f64	5458	5.458
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2875	2.875
Model recording + forward / Candle batched normalized/sliced MLP · f64	5625	5.625
Model backward / Candle prepared graph → fresh gradients · f64	20959	20.959
Model backward / Kernix prepared graph → fresh gradients · f64	3750	3.75
Parameter update / Kernix transactional SGD, fixed gradients · f64	750	0.75
Row softmax / ndarray 256 × 64 · f64	65250	65.25
Row softmax / NumPy 256 × 64 · f64	76667	76.667
Row softmax / Candle 256 × 64 · f64	64999.99999999999	64.99999999999999
Row softmax / Kernix 256 × 64 · f64	78917	78.917

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	750	0.75
Create zeros / faer 64 × 64 · f32	291	0.291
Create zeros / ndarray 64 × 64 · f32	291	0.291
Create zeros / NumPy 64 × 64 · f32	542	0.542
Create zeros / Candle 64 × 64 · f32	458	0.458
Create ones / faer 64 × 64 · f32	333	0.333
Create ones / Kernix 64 × 64 · f32	292	0.292
Create ones / Candle 64 × 64 · f32	417	0.417
Create ones / NumPy 64 × 64 · f32	1125	1.125
Create ones / ndarray 64 × 64 · f32	333	0.333